Evaluating a RAG-Based Chatbot for User-Selected Undergraduate Curriculum Comparisons at José Rizal University

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***Abstract*—This paper describes our plan to test a RAG-based chatbot that helps graduating Grade 12 students compare undergraduate programs in José Rizal University's (JRU) College of Computer Studies and Engineering (CSE). Students would pick two or more programs from a fixed, verified set of documents and ask the chatbot to compare subjects, units, and practical requirements. We want to see how the chatbot's correctness, groundedness, and completeness change as students compare more programs at once, using questions we've checked against real sources. We still need to confirm local demand, get permission for the documents we use, and verify curriculum versions, but those steps come later. Our goal is to give students reliable access to program information, which supports Sustainable Development Goal 4.**

Keywords—retrieval-augmented generation, curriculum comparison, undergraduate programs, comparison scale, SDG 4, JRU.

I. Selected UN SDG and Justification

Selected SDG: SDG 4 – Quality Education.

Relevant SDG area: Target 4.3, which concerns equal access to affordable and quality technical, vocational, and tertiary education. It calls for inclusive, equitable education and learning opportunities for everyone [1]. That fits our project, because our problem is really about students not having easy access to clear information about college programs before they even enroll.

Justification: There's research showing that when students get clear information about what a college major actually involves, it can shift their choices, especially if they're still undecided [2]. That's useful context, but it doesn't prove students at JRU actually face this problem. We still need to check that ourselves.

Connection to RAG: A RAG-based chatbot could pull real, approved curriculum documents and lay out the differences with sources attached. We're only testing whether it gives good answers. We're not claiming it will boost enrollment, make school more affordable, or move the needle on any official SDG indicator, since that's a much bigger claim than what we can actually test here.

II. Narrow Problem Statement

Picture a graduating Grade 12 student trying to compare two or more computing and engineering programs at JRU. Right now, they'd have to dig through several program pages or documents just to see which subjects overlap, how the unit counts differ, and what practical requirements each program has. That's tedious even for two programs, and it gets worse the more programs someone tries to compare at once.

This matters for two reasons. First, messy or incomplete comparisons leave students with a shaky picture of what they're signing up for. Second, RAG systems tend to get less reliable as you hand them more material to retrieve from, so it's worth actually testing how the chatbot holds up as comparisons grow. We still need to confirm, through a short needs check, that this difficulty really shows up for JRU applicants and isn't just something we assumed.

Progressive narrowing:

Level	**Group output**
Broad domain	Access to tertiary-education program information
SDG-related issue	Difficulty consolidating official curriculum information across programs
Target users	Graduating Grade 12 students considering JRU
Narrow problem	Consolidating subjects, units, and practical requirements across JRU College of Computer Studies & Engineering (CSE) undergraduate programs
Proposed chatbot focus	Source-supported comparative questions across two or more CSE programs

Boundaries: The knowledge base covers all five CSE degree programs: BS Computer Engineering (BSCpE), BS Electronics Engineering (BSEcE), BS Information Technology

(BSIT), BS Information Technology with specialization in Business Analytics (BSIT-BA), and BS Entertainment and Multimedia Computing (BSEMC, covering the Digital Animation Technology and Game Development majors). BSEMC's curriculum PDF covers both of its majors (Digital Animation Technology and Game Development) in one document; for comparison purposes, a student selecting BSEMC must also specify which major, and the chatbot will compare at the major level rather than treating BSEMC as a single undifferentiated entry. The fixed collection is therefore five degree programs, with BSEMC resolved to one of its two majors at selection time. Before we run any evaluation, we'll freeze the document collection and tag every source with its program, major, and curriculum version. We'll only include programs we can back up with a verified source, and we'll say so if something isn't supported. The comparisons themselves cover subjects, units, and practical requirements; we'll only include learning outcomes if we actually find them documented. If someone picks a lot of programs at once, we might split the answer into sections, but we won't drop any of the programs they asked about.

A few sample questions we expect: which subjects do these programs share, how do the required units differ, and how do internship requirements compare. What the chatbot won't do is just as important: it won't judge grades, tell a student which program "fits" them, guess at how hard a course is, or touch anything related to admissions. And just because two programs list a subject with a similar name doesn't mean the content is the same, so we won't treat matching titles as proof of anything. This keeps our project separate from tools that recommend a strand or judge whether someone is "ready" for a program.

III. Primary Research Question

As students compare more JRU programs at once, how much do answer correctness, groundedness, and completeness drop off, if at all, within a fixed collection of undergraduate programs?

We like this question because it's narrow: one school, one closed set of documents, one task. We can test it directly by running the same RAG setup on comparisons of increasing size, say two programs, then three, then the full set, and scoring each group on correctness, groundedness, and completeness against answers we've already checked. Unlike a simple RAG-versus-no-RAG test, we genuinely don't know the answer ahead of time. Prior work on long-context retrieval [11] suggests performance could hold steady, slide gradually, or fall off sharply past some threshold, and we want to find out which one actually happens here. To be clear, we're testing how retrieval and generation hold up under a growing comparison load, not whether a chatbot beats a plain website or table.

IV. Initial Knowledge-Base Data Sources

A quick note before we list these: they are potential sources, not a finished dataset. We haven't verified anything yet, and none of this counts as a completed research corpus. All sources below are scoped to CSE undergraduate programs, matching the fixed set described in Section II.

Field	Description
Source name/org.	JRU; five official CSE program curriculum documents [4]–[8]
Document type	Program-specific curriculum PDFs (one per BSCpE, BSEcE, BSIT, BSIT-BA, BSEMC)
Relevance	Primary source for subjects, units, and program structure per program
Availability	Publicly accessible, linked directly from the CSE program listing [3]
Reliability/permission	Official institutional origin and dated April 2024; still to be checked for applicability to incoming (vs. currently enrolled) students and for research-use permission

A. JRU College of Computer Studies & Engineering Program Listing

Producer/type: JRU; official CSE academics webpage [3]. **Relevance:** Confirms the exact set of five CSE degree programs and links directly to each program's curriculum PDF; used to build and cross-check the fixed collection. **Availability:** Public. **Reliability/permission:** Authoritative institutional origin; last modified June 2026 per page metadata, so reasonably current. On its own, the listing doesn't give subject-level detail – that comes from the per-program PDFs below.

B. CSE Program Curriculum Documents (Primary Source)

Producer/type: JRU College of Computer Studies & Engineering; five official per-program curriculum PDFs, each dated April 2024 [4]–[8]. **Relevance:** This is where we'd actually get the details on subjects, units, and practical requirements for each program – the core evidence for every comparison. **Availability:** Public, linked directly from the CSE program listing. **Reliability/permission:** Official and program-specific, more current than the general institutional overview we originally considered; still need to confirm each PDF reflects the curriculum version applicable to incoming (not currently enrolled) students, and that research use is permitted.

C. Approved Program and Subject Documents (Supplementary)

Producer/type: CSE academic units; subject descriptions and internship or program handbooks, where the program PDFs above don't fully cover practical requirements. **Relevance:** Fills gaps in practical-requirement detail not covered by the primary curriculum PDFs. **Availability:** To be requested; access is not confirmed. **Reliability/permission:** We'll only use documents where we can confirm who owns them, what version they are, and that we're allowed to use them for research. We don't need, and won't ask for, anything like student records or grades.

Source management: We'll log the program, major, curriculum year, document owner, page or link, and the date we checked it, for every CSE program in the collection. We'll

spot-check any extracted tables against the originals, gather evidence program by program rather than all at once, and flag anything missing or conflicting. Locking the source set before evaluation is what keeps our results reproducible.

V. Initial Evaluation Data Plan

Needs and question collection: We'll ask a small, informal group of graduating Grade 12 students to walk us through what they'd actually want to compare, and to try comparing programs using whatever materials exist today. We'll also ask staff or faculty what questions come up often. Participation is voluntary, and we'll follow the school's procedures for working with minors. We won't collect names or grades. Whatever we learn here is exploratory: it tells us something useful, but it isn't a formal survey result.

Preliminary question set: We're aiming for around 40 questions, checked one by one: 30 we expect to be fully answerable, five that are only partly answerable, and five where the answer just isn't in our knowledge base. That number isn't based on a statistical formula, it's just a workload we think we can manage well. Since comparison size is now what we're actually studying, we'll spread these questions across sizes, roughly even numbers of two-program, three-program, and full-set comparisons, rather than treating size as an afterthought. Within each size group, we'll mix in common subjects, unit differences, practical requirements, and programs with confusingly similar names. We'll try to work the full supported collection into the set somewhere.

Reference answers: For every question, we'll write out the correct facts and point to the exact passage that backs each one up. A faculty member or program representative will review our answer key, and two of us will independently score a sample and settle any disagreements. We'll keep the actual test questions and answers separate from anything we feed into the chatbot during development, though the source documents themselves stay in the knowledge base.

Controlled experiment: We'll lock in one model version, the same instructions, the same generation settings, and the same RAG setup for every test. The only thing that changes across conditions is comparison size, two programs, three programs, or the full set. The model is allowed to admit it doesn't know something in any condition. Where we can, we'll hide the size label from whoever's scoring, so we're not biased by knowing what we're looking at. We'll report exactly what we tested and where the limits are, rather than assuming it generalizes further than it does.

Metrics: We'll check correctness, whether the answer matches our reference facts; groundedness, whether the claims are actually backed by our approved sources; and completeness, whether the answer covers everything the question asked for. We'll also check whether citations hold up and whether the chatbot handles missing information reasonably, and we'll track how often it refuses to answer something it actually could, since we don't want to reward it for being overly cautious. We're borrowing ideas from RAGAs, which separates retrieval quality from generation quality [9], and from ALCE, which focuses on checking citations [10]. Our main scoring will be done by hand; automated scores are just a supplement.

We'll break results down by comparison size and look closely at mistakes, like mixing up which program a fact belongs to, leaving something out, or confusing curriculum versions. Bigger comparisons need real testing here, since past work on long-context models found they can lose track of information depending on where it sits in the input [11]. If we ever wanted to claim the chatbot is faster than existing materials, that would need its own separate study. We're not making that claim here.

VI. Possible Language Model and Justification

Proposed model: Qwen3-8B, running in non-thinking mode for now. It's open-weight under Apache 2.0, and the official model card lays out how to deploy and configure it [12]. Its size makes it realistic for a student project running on limited hardware, though we'll still need to check latency and whether our hardware can actually handle it. This is a starting pick, not a final answer; we haven't run comparisons against other models yet.

Expected role: We expect the model to take the retrieved passages, turn them into a clear comparison, cite where the information came from, and say so when something isn't available. Retrieval and document labeling will happen as separate steps, so we're not relying on the model itself to know JRU facts from memory. We're not planning any fine-tuning.

Practical considerations: We'll pilot this on whatever university or group computing resources we can access, and we may need a quantized version or a hosted option if our hardware falls short. How much memory we need depends on precision, context length, and the runtime we use, so we can't confirm hardware requirements yet. We'll keep the checkpoint, precision, and mode fixed across every test condition. We're starting in English; if we want to support Filipino or mixed-language input later, that needs its own set of validated questions first. We'll also track inference cost and response time during the pilot.

VII. Group Reflection on Manageability

We think this project is manageable because it's really just one task, at one university, for one group of users. Letting students flexibly pick programs doesn't turn this into personal counseling, since every answer is still tied to verified curriculum facts and a fixed set of documents. A source audit will tell us how much we can realistically cover, and keeping our test set small but well-reviewed keeps the evaluation workload reasonable. The main things we're depending on are getting current documents, confirming the problem is real at JRU, having reviewers available, and having enough computing access. If we can't cover everything, we'll just report what we did cover instead of overpromising university-wide coverage. To be clear, this proposal lays out a future study. We haven't collected data, built anything, or shown any performance improvement yet.

REFERENCES

[1] United Nations, "Goal 4." Accessed: Sep. 16, 2026. [Online]. Available: https://sdgs.un.org/goals/goal4

[2] F. Ersoy and J. D. Speer, "Opening the black box of college major choice: Evidence from an information intervention," Journal of Economic Behavior & Organization, vol. 231, art. 106800, 2025. [Online]. Available: https://doi.org/10.1016/j.jebo.2024.106800

[3] José Rizal University, "College." Accessed: Sep. 21, 2026. [Online]. Available: https://jru.edu/academics/college/

[4] José Rizal University, "Bachelor of Science in Information Technology (BSIT) Program," Apr. 2024. [Online]. Available: https://jru.edu/wp-content/uploads/2024/04/INFORMATION-TECHNOLOGY-PROGRAM.pdf

[5] José Rizal University, "Bachelor of Science in Information Technology with Specialization in Business Analytics (BSIT-BA) Program," Apr. 2024. [Online]. Available: https://jru.edu/wp-content/uploads/2024/04/INFORMATION-TECHNOLOGY-WITH-SPECIALIZATION-IN-BUSINESS.pdf

[6] José Rizal University, "Bachelor of Science in Computer Engineering (BSCpE) Program," Apr. 2024. [Online]. Available: https://jru.edu/wp-content/uploads/2024/04/COMPUTER-ENGINEERING-PROGRAM.pdf

[7] José Rizal University, "Bachelor of Science in Electronics Engineering (BSEcE) Program," Apr. 2024. [Online]. Available: https://jru.edu/wp-content/uploads/2024/04/ELECTRONICS-ENGINEERING-PROGRAM.pdf

[8] José Rizal University, "Bachelor of Science in Entertainment and Multimedia Computing (BSEMC) Program," Apr. 2024. [Online]. Available: https://jru.edu/wp-content/uploads/2024/04/ENTERTAINMENT-AND-MULTIMEDIA-COMPUTING-PROGRAM.pdf

[9] S. Es, J. James, L. Espinosa Anke, and S. Schockaert, "RAGAs: Automated evaluation of retrieval augmented generation," EACL: System Demonstrations, 2024. [Online]. Available: https://aclanthology.org/2024.eacl-demo.16/

[10] T. Gao, H. Yen, J. Yu, and D. Chen, "Enabling large language models to generate text with citations," EMNLP, 2023. [Online]. Available: https://aclanthology.org/2023.emnlp-main.398/

[11] N. F. Liu et al., "Lost in the middle: How language models use long contexts," arXiv:2307.03172, 2023. [Online]. Available: https://arxiv.org/abs/2307.03172

[12] Qwen Team, "Qwen3-8B," official model card. Accessed: Sep. 16, 2026. [Online]. Available: https://huggingface.co/Qwen/Qwen3-8B